CodeAlpha Project 1: Iris Flower Classification

Simple English • Beginner-friendly • Start-to-finish checklist

1. What is this project?

• Teach a computer to guess the species of an Iris flower from four measurements.

• Each row is one flower. The measurements are called features; the species is the answer we want the model to predict.

2. What you need

• A browser and Google Colab. Upload `CodeAlpha_Iris_Flower_Classification.ipynb` and run cells from top to bottom.

• The Iris dataset is included with scikit-learn, so you do not need to download a separate dataset.

3. Important words in simple English

• Dataset = a table of examples.

• Feature (X) = information used to make a prediction.

• Target (y) = the correct answer the model learns to predict.

• Training data = examples used to teach the model.

• Test data = examples kept aside to check how well it learned.

• Accuracy = the percentage of test examples predicted correctly.

4. Start-to-finish steps

• Step 1: Open Google Colab and upload the notebook.

• Step 2: Run the first code cell to load the data.

• Step 3: Check the number of rows, columns, missing values, and duplicate rows.

• Step 4: Look at the scatter plot. It helps you see how petal measurements differ between species.

• Step 5: Split the data into training examples and test examples.

• Step 6: Train Logistic Regression. The notebook prints test accuracy and a detailed report.

• Step 7: Read the confusion matrix. The diagonal counts correct predictions; off-diagonal values are mistakes.

• Step 8: Try a new flower. Enter four measurements in this order: sepal length, sepal width, petal length, petal width—in centimeters.

• Step 9: Write your conclusion using the results printed by your own run.

5. How to explain this in an interview

• “I built a classification model using the Iris dataset. I checked the data, explored it with a chart, split it into training and test sets, trained Logistic Regression with feature scaling, and evaluated it using accuracy, a classification report, and a confusion matrix.”

6. New idea to improve it

• Train a Decision Tree or Random Forest using the same train/test split and compare results. Explain which species each model confuses.

7. GitHub and submission

• Save the notebook as `Iris_Flower_Classification.ipynb`.

• Create a README with the project goal, dataset, steps, results, and how to run it.

• Use repository name `CodeAlpha_Iris_Flower_Classification`.

• Run every cell and check the outputs before submitting. Do not write a result you have not verified.